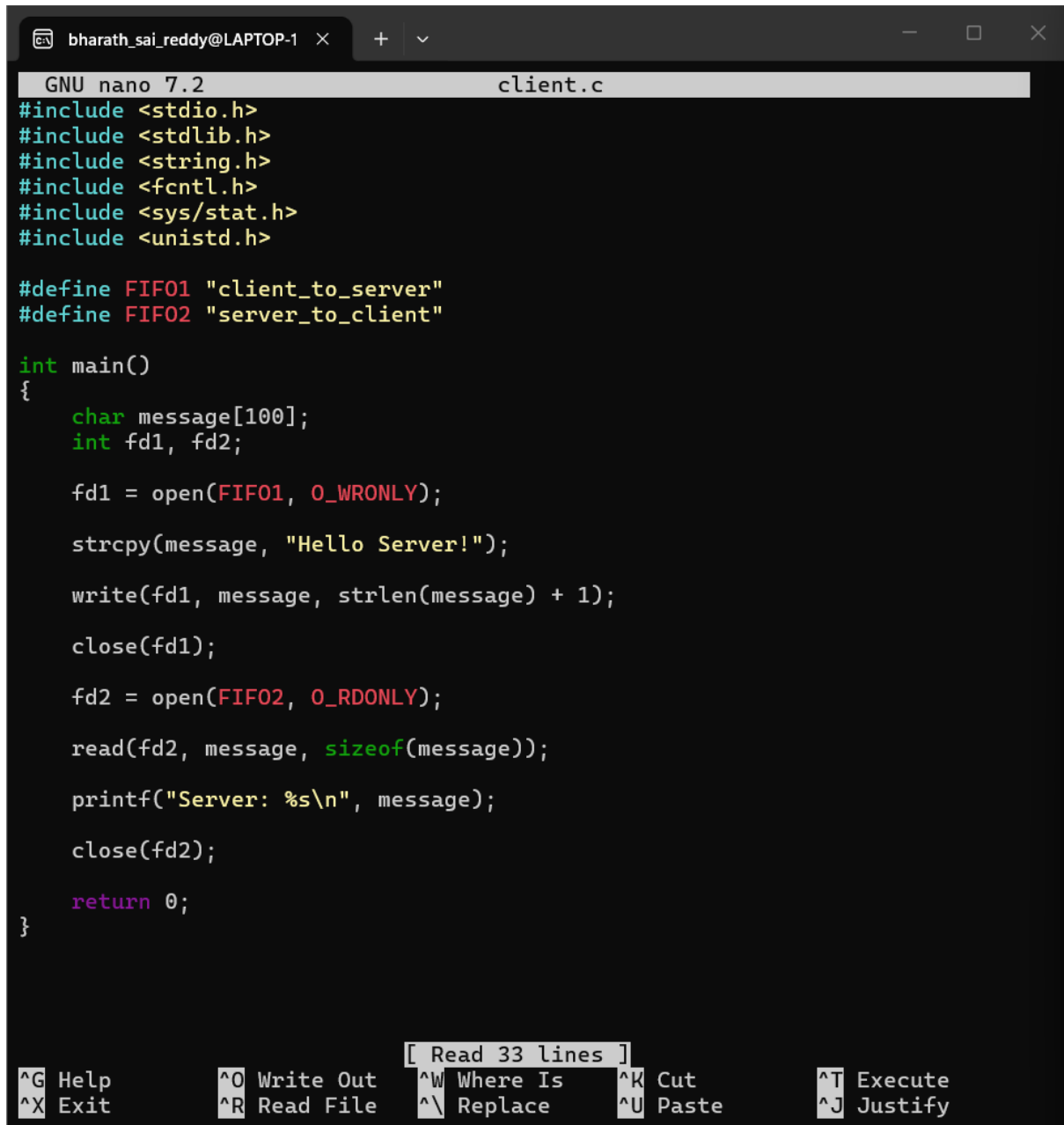


Practical -06

Server - client

Code:

Client code:



The screenshot shows a terminal window with the nano 7.2 editor open to a file named client.c. The code is written in C and uses named pipes (FIFOs) for communication. It includes headers for stdio, stdlib, string, fcntl, sys/stat, and unistd. Two FIFOs are defined: FIF01 for client-to-server communication and FIF02 for server-to-client communication. The main function opens FIF01 in write-only mode, writes the message "Hello Server!", and closes it. Then it opens FIF02 in read-only mode, reads the message from the server, and prints it with a prefix "Server: ". The status bar at the bottom shows various keyboard shortcuts and indicates that 33 lines have been read.

```
GNU nano 7.2 client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIF01 "client_to_server"
#define FIF02 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

    fd1 = open(FIF01, O_WRONLY);

    strcpy(message, "Hello Server!");

    write(fd1, message, strlen(message) + 1);

    close(fd1);

    fd2 = open(FIF02, O_RDONLY);

    read(fd2, message, sizeof(message));

    printf("Server: %s\n", message);

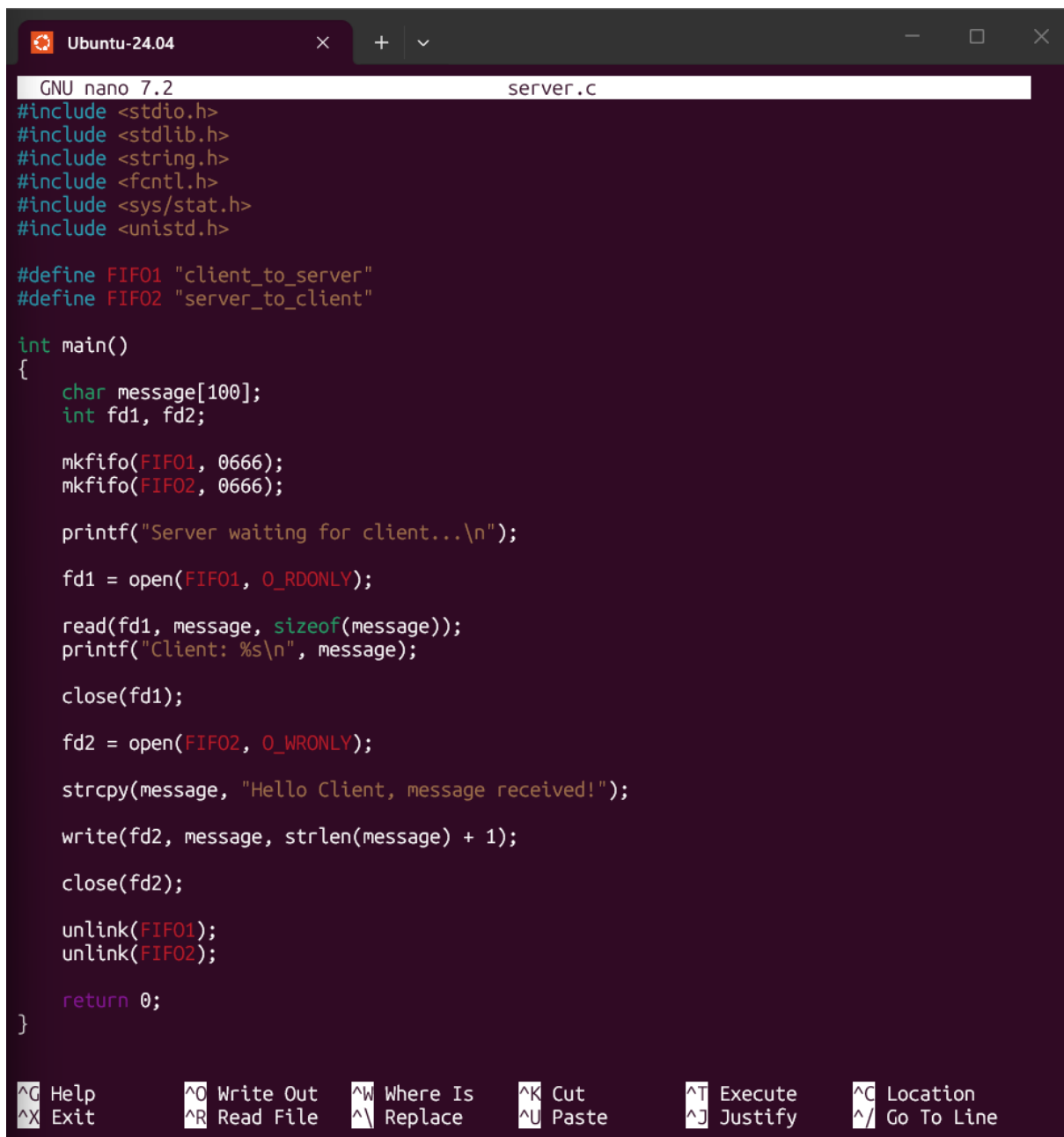
    close(fd2);

    return 0;
}
```

[Read 33 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

Server Code:



```
GNU nano 7.2 server.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main()
{
    char message[100];
    int fd1, fd2;

    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server waiting for client...\n");

    fd1 = open(FIFO1, O_RDONLY);

    read(fd1, message, sizeof(message));
    printf("Client: %s\n", message);

    close(fd1);

    fd2 = open(FIFO2, O_WRONLY);

    strcpy(message, "Hello Client, message received!");

    write(fd2, message, strlen(message) + 1);

    close(fd2);

    unlink(FIFO1);
    unlink(FIFO2);

    return 0;
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line

OutPut:

```
bharath_sai_reddy@LAPTOP-1  x  +  v  -  □  x
System load:  0.56      Processes:           67
Usage of /:   0.3% of 1006.85GB    Users logged in:      0
Memory usage: 6%      IPv4 address for eth0: 172.29.142.237
Swap usage:   0%

* Canonical Workshop gives developers fast, composable, reproducible, and
secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/home/bharath_sai_reddy/.hushlogin file.
bharath_sai_reddy@LAPTOP-1711DF88:~$ ls
90128_OSSP  OSLab  filedescriptor  markdown-project
bharath_sai_reddy@LAPTOP-1711DF88:~$ cd 90128_OSSP
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ ls
2520090128_Practical  2520090128_Skill  File_Permission  ForgeOS  Scheduling
algorithms  filedescriptor  pipes  signals
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ cd 2520090128_Practical
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ ls
Practical-01  Practical-03  Practical-05  Practical-07  Practical-09
Practical-02  Practical-04  Practical-06  Practical-08  Practical-10
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd Prac
tical-06
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ls
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ nano server.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ nano client.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ gcc server.c -o server
gcc client.c -o client
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ./server
Server waiting for client...
Client: Hello Server!
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ |

Ubuntu-24.04  x  +  v  -  □  x
bharath_sai_reddy@LAPTOP-1711DF88:~$ ls
90128_OSSP  OSLab  filedescriptor  markdown-project
bharath_sai_reddy@LAPTOP-1711DF88:~$ cd 90128_OSSP
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ ls
2520090128_Practical  2520090128_Skill  File_Permission  Schedulingalgorithms  pipes
2520090128_Skill  ForgeOS  filedescriptor  signals
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP$ cd 2520090128_Practical
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ ls
Practical-01  Practical-03  Practical-05  Practical-07  Practical-09
Practical-02  Practical-04  Practical-06  Practical-08  Practical-10
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd Prac
tical-06
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$ ./client
Server: Hello Client, message received!
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practica
l-06$
```

Signals

Code:

```
Ubuntu-24.04 × + ▾
GNU nano 7.2 signals.c
#include <stdio.h>
#include <stdlib.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received!\n");

    else if (sig == SIGTERM)
        printf("SIGTERM received!\n");

    else if (sig == SIGUSR1)
        printf("SIGUSR1 received!\n");
}

int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);
    signal(SIGUSR1, handle_signal);

    printf("Process ID: %d\n", getpid());
    printf("Waiting for signals...\n");

    while (1)
    {
        sleep(1);
    }

    return 0;
}
```

Output:

```
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Skill$ cd 2520090128_Skill
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Skill$ ls
Skill-01 Skill-02 Skill-03 Skill-04 Skill-05 Skill-06 Skill-07 Skill-08 Skill-09 Skill-10
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Skill$ cd Skill-03
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Skill/Skill-03$ cd ..
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Skill$ cd ..
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP$ ls
2520090128_Practical 2520090128_Skill File_Permission ForgeOS Schedulingalgorithms file
descriptor pipes signals
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP$ cd 2520090128_Practical
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical$ ls
Practical-01 Practical-02 Practical-03 Practical-04 Practical-05 Practical-06 Practic
al-07 Practical-08 Practical-09 Practical-10
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical$ cd Practical-06
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ nano sign
als.c
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ gcc signa
ls.c -o signals
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ ./signals
Process ID: 993
Waiting for signals...
^C
SIGINT received!
SIGUSR1 received!
SIGTERM received!
Killed
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$
```

```
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ kill -SIGUSR1 12345
-bash: kill: (12345) - No such process
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ kill -SIGUSR1 993
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ kill -SIGTERM 993
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$ kill -9 993
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-06$
```